Synergistic toxicity of the environmental neurotoxins methylmercury and β-N-methylamino-L-alanine.

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Determination of the environmental factors involved in neurodegenerative diseases has been elusive. Methylmercury and β-N-methylamino-L-alanine (BMAA) have both been implicated in this role. Exposure of primary cortical cultures to these compounds independently induced concentration-dependent neurotoxicity. Importantly, concentrations of BMAA (10-100 μM) that caused no toxicity alone potentiated methylmercury (3 μM) toxicity. In addition, concentrations of BMAA and methylmercury that had no effect by themselves on the main cellular antioxidant glutathione together decreased glutathione levels. Furthermore, the combined toxicity of methylmercury and BMAA was attenuated by the cell permeant form of glutathione, glutathione monoethyl ester. The results indicate a synergistic toxic effect of the environmental neurotoxins BMAA and methylmercury, and that the interaction is at the level of glutathione depletion.